BAMBINOS.LIVE • OPERATIONS EXCELLENCE CASE

Re-Engineering & Automating the Student Renewal → Class Scheduling Pipeline

A Complete Zero-Touch Workflow Architecture, Empirical Leakage Audit, Working Reconciliation Engine, Interactive Operations Dashboard & 30-60-90 Day Scaling Plan

Role Evaluated:	Senior AI Program Manager (Operations)
Hiring Assessment:	Stage 2 — Live Operations Case Take-Home
Focus Areas:	Process Automation • Exception Triage • Operational SLAs • No-Code Architecture
Dataset Evaluated:	Official 5-Table Operations Dataset (./ops/data/)
Prototype Repository:	https://github.com/ankitsingh9162/Bambinos-AI-Program-Manager
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Operational Assessment Mandate: *Re-engineer the student renewal pipeline from parent payment completed to class sessions booked. Eliminate repetitive administrative overhead for 200+ teachers, empower a 10-person Operations team to scale smoothly, and provide executive transparency through real-time operational control.*

Table of Contents & Executive Summary

Table of Contents:

1. Executive Summary & Core Results	Page 2
2. Business Context & The Legacy 4-Table Pipeline	Page 3
3. Deliverable A — Quantitative Funnel & Root-Cause Audit	Page 4
4. Student-by-Student Ground-Truth Reconciliation Matrix	Page 5
5. Deliverable B — Re-Engineered Zero-Touch Architecture	Page 6
6. Ten Standardized Reason Codes & Safeguards	Page 7
7. Deliverable C — Prototype Reconciliation Engine & Results	Page 8
8. Deliverable D — Executive Dashboard & Daily Action Table	Page 9
9. Standard Operating Procedure (SOP) & 10-Person RACI	Page 10
10. 30-60-90 Day Implementation Roadmap & Leadership Decisions	Page 11

Executive Summary:

Student renewals drive the core retention and high-LTV organic revenue at **Bambinos.live**. However, an audit of the production dataset across the four-table pipeline (*1_invoices* → *2_faculty_form_unapproved* → *3_booking_requests* → *4_class_schedule_bookings*) reveals an alarming operational breakdown: out of **13 generated invoices** (11 Paid, 1 Partially Paid, 1 Failed), **only 1 single student (68101)** was successfully scheduled into live classes. This represents an end-to-end operational conversion rate of just **7.7%** (or **9.1%** of paid renewals). Over **90.9% of paying parents** face administrative delays, unbooked sessions, or operational silence.

The Core Structural Flaw: The current architecture treats teachers as administrative data clerks and Ops executives as human middleware. Teachers pitch renewals during classes, but making them manually submit Google Forms after payments results in **18.2% Orphaned Paid Invoices** (INR 29,000 in collected fees left in scheduling limbo for Students 68106 and 68110). Furthermore, Ops manually reviews every clean renewal, causing an unapproved backlog where **61.5% of the student cohort** hits friction (phone mismatches, batch overcapacity, typos, and browser duplicates).

The Transformation Blueprint: We propose inverting the operational model from a manual 'push-by-teacher' to an automated 'event-driven pull'. A payment gateway webhook immediately triggers student verification from master reference tables. Clean renewals bypass manual review entirely via a **Zero-Touch Auto-Approval Engine**, while edge cases are routed to a 10-person Ops team using standardized reason codes, interactive WhatsApp self-service slot pickers, and a live Streamlit Operations Head Dashboard. In prototype testing, this solution achieved a **45.5% auto-approval yield on day one** with 100% unique booking IDs, zero financial leaks, and a verified 20-test automated validation suite.

Metric	Legacy Baseline (Observed)	Prototype Result	Proposed Target SLA
End-to-End Scheduling Conversion	7.7% (1 / 13 Students)	45.5% Auto-Approved (5/11 Paid)	>= 80.0% Clean Auto-Approval
Orphaned Paid Invoice Rate	18.2% (2 / 11 Paid; INR 29k at risk)	0% Orphaned in Auto-Flow	0.0% (< 2h Automated Alert)
Happy-Path Scheduling TAT	104.7 Mins (Single student 68101)	< 5 Seconds (Simulated)	< 15 Minutes End-to-End
Automated Test Suite Status	No Automated Tests	20 Passed, 0 Failed (0.15s)	100% Regression Coverage
Overnight Pending Backlog	11 Students Awaiting Schedule	8 Unresolved (Triage Assigned)	0 Students Overnight Backlog

Business Context & The Legacy 4-Table Flow

Bambinos.live conducts thousands of live, small-group online classes weekly for children aged 4–16 across India, GCC, and the US/UK. Because live teaching builds strong interpersonal trust, the child's current teacher pitches term renewals directly to parents near the completion of a package. However, translating that completed payment into live calendar bookings currently depends on four disconnected tables/sheets with manual handoffs:

Stage & Table	How It Works Today (Legacy)	Human Owner	Vulnerability & Failure Mode
Step 1: *1_invoices.csv*	Parent pays via gateway link. A row is inserted in *1_invoices*.	Parent / Payment Gateway	Invoices settle silently. System does not automatically notify Operations or schedule classes.
Step 2: *2_faculty_form_unapproved*	Teacher manually fills Google Form, typing student details, invoice link, and fee.	Teacher (Faculty)	**Major Leakage Point:** Teachers forget to submit forms (18.2% orphaned rate), make invoice typos, or resubmit on browser freeze.
Step 3: *3_booking_requests.csv*	Ops team manually inspects each form, cross-verifies gateway payment, and creates a booking row.	Operations Team	**Bottleneck:** 8 of 12 forms sit unapproved. Clean renewals wait hours in queue. Contact mismatches halt review.
Step 4: *4_class_schedule_bookings*	From booking request, Ops or parent selects batch/time and generates class calendar rows.	Parent / Operations Team	**Drop-Off Point:** Batch overcapacity (6/6 full) causes abrupt failure. Manual slot coordination takes days over phone/chat.

The Human Bottleneck Illustrated:

Under this 4-step chain, every single renewal—regardless of how clean or standard—requires human intervention at three separate boundaries: Parent → Teacher, Teacher → Ops, and Ops → Calendar. When renewal volumes scale from tens to hundreds weekly across 200+ teachers, the 10-person Ops team is overwhelmed by repetitive verification, while paying parents experience class interruptions and frustration.

Deliverable A: Quantitative Leakage Funnel & Root-Cause Audit

A mathematical cross-table reconciliation of all records in ops/data/ was conducted to establish exact cohort metrics. The preliminary observation of *13 Invoices → 12 Forms → 4 Bookings → 1 Scheduled* was strictly verified at the unique student and invoice level:

Pipeline Stage	CSV Rows	Unique Students	Unique Invoices	Stage Conversion	Cumulative Conversion	Students Lost / Reason
1. Invoices Generated	13	13	13	100.0%	100.0%	Baseline (11 Paid, 1 Partial, 1 Failed)
2. Faculty Forms Submitted	12	11	10 direct / 11 logical	84.6%	84.6%	2 Orphaned Invoices (84906, 84910); 1 Browser Duplicate (3309)
3. Booking Requests Created	4	4	4	36.4%	30.8%	7 Students stuck in unapproved Ops review (3303, 3304, 3307, etc.)
4. Class Schedule Bookings	2	1	1	25.0%	**7.7%**	3 Bookings unfulfilled (5502 pending, 5505 full, 5512 change)

Primary Root Causes of Operational Breakdown:

- **Orphaned Paid Invoices (18.2% of Paid):** Invoices 84906 (INR 18,000) and 84910 (INR 11,000) settled successfully at the gateway, but teachers never submitted forms. No classes were scheduled, creating severe parent churn and refund risk.
- **Teacher Key Typos:** Form 3313 entered invoice 84931 instead of 84913. Unrestricted free-text form inputs break database joins, requiring manual Ops cross-referencing.
- **Unverified Payment Proof:** Form 3311 was submitted based on an unverified UPI screenshot, but Invoice 84911 failed at the gateway. Manual forms risk giving free classes to failed transactions.
- **Batch Overcapacity:** Booking 5505 pitched Student 68105 into Batch 9205, which already held 6/6 active students, causing an abrupt failure without an automatic sister batch alternative.
- **Contact Identity Mismatches:** Form 3304 (Student 68104) had a payer phone (...XXXX77) differing from the registered student mobile (...XXXX04), freezing Ops review.
- **Severe Hand-off Turnaround:** For the single student scheduled (68101), end-to-end turnaround took **104.7 minutes** across manual handoffs. All other 12 students remain indefinite (> 48h).

Cohort Master Reconciliation Matrix (All 13 Students)

This master table provides the audited, ground-truth operational trace for all 13 students across all 5 CSV datasets, linking payments, teacher forms, booking requests, and calendar schedules with exact root causes:

Student ID	Payer Name & Contact	Invoice (Net INR)	Payment Status	Faculty Form & Status	Booking & Status	Scheduled	Operational Triage Category & Action
68101	Pr***a Sh***a +91-9811XXXX01	84901 (12,000)	Paid	3301 (Approved)	5501 (Completed)	Yes (2)	**Clean Renewal (Happy Path):** Auto-approved. Reused existing completed booking 5501.
68102	Ka***k Na***r +91-9822XXXX02	84902 (10,500)	Paid	3302 (Approved)	5502 (Pending)	No	**Clean Renewal (Happy Path):** Auto-approved. Reused existing pending booking 5502.
68103	Ne***a Ve***a +91-9833XXXX03	84903 (9,000)	Paid	3303 (Pending)	None	No	**Clean Renewal (Happy Path):** Auto-approved. Created simulated booking 5601.
68104	Ri***u Iy***r +91-9844XXXX77	84904 (12,000)	Paid	3304 (Pending)	None	No	**PHONE_MISMATCH:** Payer phone differs from registered profile (...04). Ops verify.
68105	Me***a Pa***l +91-9855XXXX05	84905 (11,000)	Paid	3305 (Approved)	5505 (Batch_Full)	No	**BATCH_FULL:** Batch 9205 is 6/6 full. Route to sister batch with open seats.
68106	Fa***n Kh***n +971-5012XXX06	84906 (18,000)	Paid	None (Missing)	None	No	**FACULTY_FORM_MISSING:** Orphaned invoice. Teacher prompt via WhatsApp.
68107	Po***a Jo***i +91-9877XXXX07	84907 (7,600)	Paid	3307 (Pending)	None	No	**SCHOLARSHIP_APPROVAL_REQ:** 20% discount applied. Ops Head sign-off.
68108	Ro***n Ma***a +1-40855XXX08	84908 (22,000)	Paid	3308 & 3309	None	No	**HAPPY PATH (+ Warning):** Form 3309 merged as resolved duplicate; 3308 approved (5602).
68109	Sn***a Ku***i +91-9899XXXX09	84909 (5,250)	Partial	3310 (Pending)	None	No	**PARTIAL_PAYMENT:** Paid 5,250 of 10,500. Hold schedule; dispatch Tranche 2 link.
68110	Vi***m Re***y +91-9800XXXX10	84910 (11,000)	Paid	None (Missing)	None	No	**FACULTY_FORM_MISSING:** Orphaned invoice. Prompt educator Vikram via WhatsApp.
68111	An***i Gu***a +91-9811XXXX11	84911 (12,000)	Failed	3311 (Pending)	None	No	**PAYMENT_FAILED:** Gateway failed. Block scheduling; dispatch retry link.
68112	De***k Me***n +91-9822XXXX12	84912 (9,000)	Paid	3312 (Approved)	5512 (Awaiting)	No	**BATCH_CHANGE_REQUIRED:** Weekend slot requested. Interactive slot picker.
68113	De***a Me***a +91-9833XXXX13	84913 (11,500)	Paid	3313 (Pending)	None	No	**HAPPY PATH (+ Warning):** Invoice typo 84931 auto-matched to 84913; approved (5603).

Deliverable B: Re-Engineered Zero-Touch Architecture

To break the operational bottleneck, we redesign the renewal pipeline from a manual 'push-by-human' model to an automated 'event-driven pull' model. The teacher form is completely eliminated as a mandatory gatekeeper for standard renewals:

LEGACY REVERSE MODEL (High Friction & Leakage):
Parent Pays → Teacher Must Remember to Fill Google Form → Ops Manually Cross-Checks Gateway → Ops Allocates Batch → Calendar Booked
Result: 18.2% Orphaned Invoices, 61.5% Exception Backlog, 104.7 min Turnaround Time.

RE-ENGINEERED ZERO-TOUCH MODEL (Event-Driven Stream):
Payment Gateway Webhook (`status = Paid`) → Ingest & Pull Student Master Roster → Automated Integrity & Capacity Validation →
[Branch A: Happy Path] Instant Zero-Touch Class Scheduling (< 15 mins) + Automated WhatsApp Calendar Invites.
[Branch B: Exception Triage] Standardized Reason Code Tagging → Targeted Routing to Ops, Parent, or Teacher.

Recommended Technology Stack:

Layer	Recommended Tool	Operational Role & Strategic Rationale
Workflow Engine	n8n (Self-Hosted) / Make	Low-latency webhook ingestion, multi-branch deterministic routing, and native database connectors without per-task execution fees.
Operational Store	PostgreSQL / Supabase	ACID-compliant relational database. Enforces row-level atomic locks (WHERE active_count < max_size) to eliminate race conditions.
Ops Dashboard	Streamlit / Retool	Executive KPI tracking, queue aging alerts, and 1-click manual override console for the 10-person Ops team.
Messaging Gateway	WhatsApp Cloud API	Interactive slot pickers for parent batch changes, 1-tap confirmation prompts for teachers, and calendar invites.
Internal Alerts	Slack API	Real-time notifications for P0 payment failures and 1-click discount approval cards in #ops-approvals.
AI / NLP Layer	Claude 3.5 Sonnet / GPT-4o	Semantic parsing of unstructured remarks and personalized message drafting. **Zero authority over financial approvals.**

Five Non-Negotiable Automation Safeguards:

1. **Mandatory Financial Ground Truth:** No class booking may ever be generated without a verified, settled payment gateway transaction ID. Teacher text claims and parent payment screenshots carry zero booking privileges.
2. **Zero LLM Financial Authority:** Generative AI models are strictly constrained to text parsing and draft composition. LLMs cannot modify invoice statuses, authorize discounts, or alter batch allocations.
3. **Deterministic Idempotency:** Every pipeline transaction enforces composite keys (SHA256(student_id + invoice_id)) to prevent duplicate bookings or double billing.
4. **Hard Atomic Capacity Locks:** Batch capacity (6/6 ceiling) is locked at the database query level to prevent concurrent over-enrollment.
5. **Immutable Audit Logging:** Every state transition is written to an append-only audit trail with a 1-click manual override switch for Ops leadership.

Ten Standardized Reason Codes & Triage Schema

To eliminate ambiguous status descriptions across spreadsheets, all transactions are classified into **1 Happy-Path Code** and **9 Standardized Exception Codes**:

Reason Code	Category	Detection Condition	Automated Action	Owner & Target SLA
1. HAPPY_PATH_AUTO_APPROVED	Happy Path	Full payment settled + same batch + capacity open + contact matches.	Auto-generate booking and schedule 24 sessions; dispatch calendar WhatsApp.	Automated Engine Target: < 15 Mins
2. FACULTY_FORM_MISSING	Omission	Paid invoice > 2h with no teacher form in database (Invoices 84906, 84910).	Pull student master roster; initiate renewal; send 1-tap WhatsApp nudge to teacher.	Ops Tier 1 / Teacher Target: < 2 Hours
3. PHONE_MISMATCH	Discrepancy	Payer mobile differs from registered profile (Student 68104).	Query secondary family contacts in master user profile; route to Ops queue.	Ops Tier 1 Agent Target: < 2 Hours
4. PARTIAL_PAYMENT	Financial	Payment status is Partially_Paid or net amount < term fee (Invoice 84909).	Hold scheduling; calculate balance (INR 5,250); generate Tranche 2 link to parent.	Finance / Retention Lead Target: < 4 Hours
5. PAYMENT_FAILED	Financial	Payment status is Failed at gateway (Invoice 84911).	Reject unverified submission; block scheduling; dispatch retry link to parent.	Gateway / Ops Tier 1 Target: < 15 Mins
6. INVOICE_ID_MISMATCH	Discrepancy	Form invoice ID not found in invoice table (Form 3313 entered 84931).	Auto-reconcile if single student invoice match (warning); escalate if ambiguous.	Reconciliation Engine Target: < 30 Mins
7. DUPLICATE_FORM	Technical	Multiple forms for same student & invoice (Forms 3308 & 3309).	Merge identical duplicate as resolved warning; flag conflicting duplicate to Ops.	Automated Engine Target: < 5 Mins
8. BATCH_FULL	Capacity	Target batch occupancy == max capacity (6/6) (Booking 5505 / Student 68105).	Hold booking; query sister batches with open seats; present 2 slots to parent.	Academic Ops Lead Target: < 3 Hours
9. BATCH_CHANGE_REQUIRED	Scheduling	Parent requested slot/day change (Booking 5512 / Student 68112).	Hold booking; dispatch interactive WhatsApp slot picker widget showing open slots.	Parent Self-Service Target: < 4 Hours
10. SCHOLARSHIP_APPROVAL_REQ	Authorization	Custom discount applied in form/invoice (Form 3307 / Student 68107).	Verify tenure; post approval card to Slack #ops-approvals with 1-click buttons.	Head of Operations Target: < 2 Hours

Deliverable C: Working Prototype & Verification

A working Python prototype (ops/engine/reconciliation_engine.py) was developed to validate the proposed logic. The engine ingests all five CSV datasets in read-only mode, enforces deterministic validation rules, generates output CSV/JSON files, and is covered by an automated test suite (ops/engine/tests/test_reconciliation.py).

Empirical Reconciliation Results (Sample Cohort):

Metric / Processing Counter	Count	Verification Status & Details
Total Invoices Ingested	13	11 Paid (INR 140.1k), 1 Partially Paid (INR 5.25k), 1 Failed (INR 12k).
Unique Students Processed	13	Evaluated across 12 faculty form submissions and reference roster.
Unique Renewals Auto-Approved	5	Students 68101, 68102, 68103, 68108, 68113. **45.5% Paid Auto-Approval Yield.**
Existing Bookings Reused	2	Booking 5501 (Completed) & Booking 5502 (Pending) reused; zero duplicate creation.
New Simulated Bookings Created	3	Unique IDs assigned in 5600+ namespace: 5601, 5602, 5603. Zero ID collisions.
Unresolved Blocking Exceptions	8	Students 68104, 68105, 68106, 68107, 68109, 68110, 68111, 68112 (61.5% cohort rate).
Resolved Non-Blocking Warnings	2	Student 68108 (duplicate form merged) and 68113 (invoice typo auto-matched).
Overlap (Approved vs Unresolved)	**0**	**Strict mathematical partition:** 5 Auto-Approved + 8 Unresolved = Exactly 13.
Notification Drafts Generated	27	Personalized message copy pre-drafted for Parents, Teachers, and Ops.

Automated Test Suite Verification (20 Tests Passed):

The test suite was executed via python -m unittest ops/engine/tests/test_reconciliation.py. All 20 unit and integration tests passed cleanly in 0.157 seconds. Key test coverage includes:

- **Booking Reuse Integrity:** Verified that completed booking 5501 and pending booking 5502 are reused without generating redundant booking IDs.
- **Financial Failure Safeguard:** Confirmed that partial payments (68109) and failed payments (68111) are strictly blocked from auto-approval.
- **Conflicting Duplicate Handling:** Verified that duplicate forms with conflicting amounts trigger human review.
- **Ambiguous Typo Escalation:** Verified that typos with zero or multiple matches escalate to Ops Tier 1.
- **Zero-Overlap Partition Assertion:** Confirmed that the set intersection between auto-approved students and unresolved exception students is strictly empty.

Deliverable D: Operations Dashboard & Action Table

To provide executive visibility and daily workflow governance, a dedicated Streamlit dashboard (ops/dashboard/app.py) was constructed. It loads the audited output files and presents real-time KPI cards, funnel stages, batch occupancy, and a prioritized daily action table:

Daily Operations Action Table (Cohort Prioritization):

Student ID	Invoice	Reason Code	Pipeline Status	Next Required Operational Action	Owner & Priority
68111	84911	PAYMENT_FAILED	FLAGGED_TRIAGE	Payment failed at gateway. Dispatch retry link; alert educator classes cannot start.	Gateway / Ops T1 **P0 (Critical)**
68105	84905	BATCH_FULL	FLAGGED_TRIAGE	Batch 9205 is 6/6 full. Offer 2 sister batches to parent via interactive WhatsApp.	Academic Ops T2 **P1 (High)**
68106	84906	FACULTY_FORM_MISSING	FLAGGED_TRIAGE	Paid INR 18,000. Teacher omitted form. Dispatch 1-tap WhatsApp nudge to educator.	Ops Tier 1 / Teacher **P1 (High)**
68109	84909	PARTIAL_PAYMENT	FLAGGED_TRIAGE	Net paid INR 5,250 of 10,500. Hold classes; dispatch Tranche 2 balance link.	Finance / Retention **P1 (High)**
68110	84910	FACULTY_FORM_MISSING	FLAGGED_TRIAGE	Paid INR 11,000. Teacher omitted form. Dispatch 1-tap WhatsApp nudge to educator.	Ops Tier 1 / Teacher **P1 (High)**
68104	84904	PHONE_MISMATCH	FLAGGED_TRIAGE	Payer mobile (...77) differs from student (...04). Verify family link in profile.	Ops Tier 1 **P2 (Medium)**
68107	84907	SCHOLARSHIP_APPROVAL_REQ	FLAGGED_TRIAGE	20% discount applied. 1-click authorization card posted to Slack #ops-approvals.	Head of Operations **P2 (Medium)**
68112	84912	BATCH_CHANGE_REQUIRED	FLAGGED_TRIAGE	Weekend slot requested. Dispatch interactive WhatsApp slot picker widget to parent.	Parent Self-Service **P2 (Medium)**
68101-03, 08, 13	Various	HAPPY_PATH_AUTO_APPROVED	AUTO_APPROVED	Clean renewals fully scheduled. Automated WhatsApp confirmation sent.	Automated Engine **P3 (Completed)**

Batch Capacity Roster (Snapshot from Reference Data):

The dashboard actively tracks occupancy across batches, highlighting full and near-capacity classes: **Batch 9205 (CM-G4-Batch-D)** is full at 6/6 capacity; **Batches 9202, 9206, 9208, 9210, 9211** are near capacity (5/6); and **Batches 9201, 9203, 9207, 9209, 9213** have open capacity (3/6 or 4/6). Enforcing atomic database capacity limits prevents over-enrollment while maintaining teacher-to-student quality ratios.

Standard Operating Procedure & Team RACI Matrix

To support **200+ teachers** across thousands of weekly renewals with a **10-person Operations team**, Ops members must not spend time manually transcribing data. Operations shifts entirely to an exception-handling model:

10-Person Operations Team Allocation:

Role	Headcount	Core Operational Responsibilities
Operations Head	1	Executive pipeline governance, SLA compliance review, policy sign-offs, and commercial discount approvals.
Operations Team Lead	1	Daily stand-up coordinator, queue assignment, P0/P1 exception escalation manager, shift coverage.
Tier-1 Ops Executives	4	Phone mismatch verification, ambiguous typo investigation, teacher 1-tap reminders, parent slot chasing.
Tier-2 Ops Executives	2	Academic batch management, sister batch reassignments, educator calendar coordination, capacity rebalancing.
Retention Specialist	1	Partial payment tranche collections, custom retention discount evaluation, parent cancellation salvage.
Systems & Data Specialist	1	n8n workflow monitoring, webhook health, database concurrency locks, dashboard maintenance, audit logs.

Operations RACI Matrix:

Pipeline Activity	Ops Head	Team Lead	Tier 1	Tier 2	Teacher	Finance	Parent
Payment Gateway Ingestion	I	I	I	I	I	A	C
Zero-Touch Auto-Approval	A	I	I	I	I	I	I
Orphaned Invoice Follow-Up	A	C	R	I	R	I	C
Contact Mismatch Verification	A	I	R	I	C	I	C
Partial Payment Chasing	A	I	C	I	C	R	R
Failed Payment Resolution	A	I	R	I	I	C	R
Batch Capacity Reallocation	A	C	I	R	C	I	C
Scholarship Sign-Off	A / R	C	C	I	C	C	I
Calendar Session Booking	A	I	I	R	I	I	I

R = Responsible (Executes), A = Accountable (Final Sign-off), C = Consulted (Input), I = Informed (Updated)

30-60-90 Day Roadmap & Leadership Decisions

A controlled, three-phase rollout ensures that automation is introduced safely without disrupting live student classes:

Phase & Timeline	Core Objectives & Scope	Key Operational Deliverables	Proposed Target Metrics
Days 0–30: Foundation & Control	Standardize 10 reason codes; deploy prototype dashboard; audit webhook feasibility; conduct shadow simulation on 100 historical renewals.	SOP v1.0, Reason Code Catalog, Staging n8n Engine, Dashboard v1, Pilot Readiness Checklist.	>= 99% shadow engine accuracy; zero schema errors.
Days 31–60: Automation Pilot	Activate live webhook ingestion and auto-scheduling for a controlled pilot cohort of 25 teachers (~250 renewals/week planning estimate).	Live Webhook Integration, Human-in-the-Loop Triage, WhatsApp Interactive Slot Picker, Audit Logging.	>= 75% pilot auto-approval yield; < 15 min clean TAT; 0 duplicate bookings.
Days 61–90: Scale & Optimize	Scale automation to all 200+ teachers; transition 10-person Ops team to exception-only triage; enforce zero overnight backlog.	SOP v2.0 (Scaled), Full Faculty Cutover, Daily Automated Executive Health Digest, Q2 Optimization Backlog.	**>= 80% Company-Wide Auto-Approval**; 0.0% Orphaned Rate; 0 Overnight Backlog.

Formal Decisions Required from Operations Leadership:

1. **Batch Capacity Cap Policy:** Formally confirm whether 6 students is a hard immutable cap or if temporary +1 expansion (max 7) is permissible with manager approval.
2. **Tiered Scholarship Authorization:** Approve the proposed authorization tiers (Discounts <= 10% auto-allowed for tenure > 6m; 11–20% requires Ops Lead; > 20% requires Ops Head).
3. **Teacher Attribution Model:** Confirm transition from manual Google Forms to automated reference roster mapping, protecting teacher renewal commissions while eliminating orphaned invoice delays.
4. **Safe Typo Auto-Correction Policy:** Authorize automated reconciliation of single-student invoice typos (such as Form 3313) while mandating human review for ambiguous cases.

Conclusion & Strategic Recommendation:

By replacing the legacy 4-table manual handoff process with the proposed Zero-Touch Reconciliation Engine, Bambinos.live can eliminate 18.2% orphaned invoice leakage, reduce payment-to-schedule turnaround from 104+ minutes to under 15 minutes, and empower a 10-person Operations team to scale smoothly across 200+ teachers. The prototype is verified, tested across 20 automated scenarios, and ready for Phase 1 operational rollout.